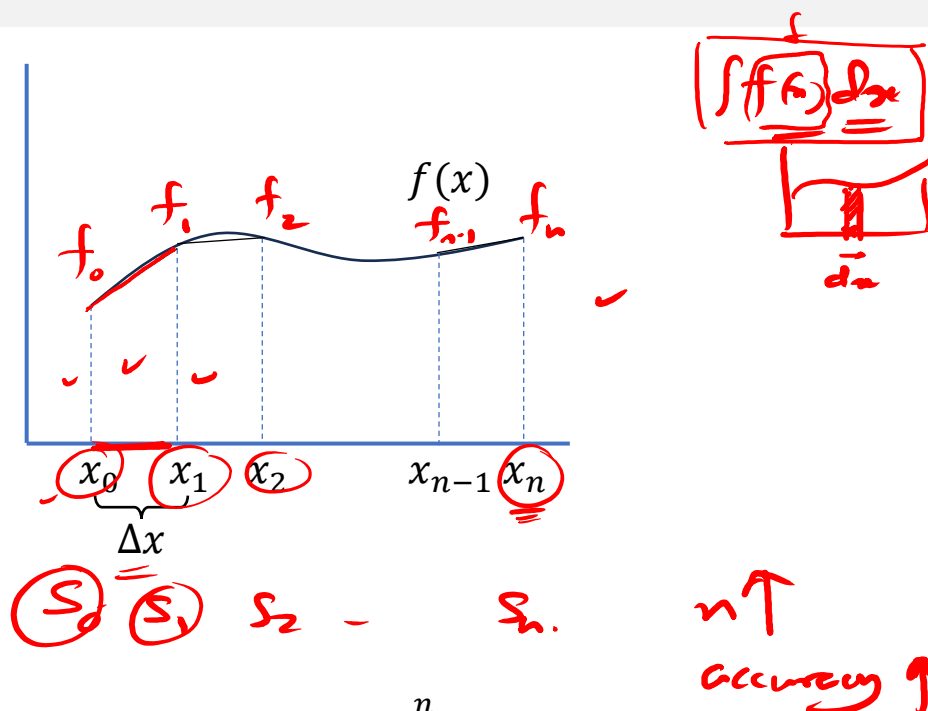


Option Pricing using Fourier Transform (2. Quadrature, DFT, FFT)

Numerical Integration (Recap)

Numerical integration can be used to solve the area under a curve. The above implementation is for a specific technique called as trapezoidal rule where the area is broken into small trapeziums and then the total area is the discrete sum of these areas.

$$\begin{aligned}
 & \left(\frac{1}{2} \Delta x \cdot (f_0 + f_1) \right) + \left(\frac{1}{2} \Delta x (f_1 + f_2) \right) + \\
 & \quad \dots + \left(\frac{1}{2} \Delta x (f_{n-1} + f_n) \right) \\
 & = \Delta x \sum f_j w_j
 \end{aligned}$$



$$\text{Area} = \int_{x_0}^{x_n} f(x) dx \approx \sum_{j=0}^n w_j f_j \Delta x$$

$$w_j = \begin{cases} 0.5, & j = 0, n \\ 1, & \text{otherwise} \end{cases}$$

Trapezoidal rule

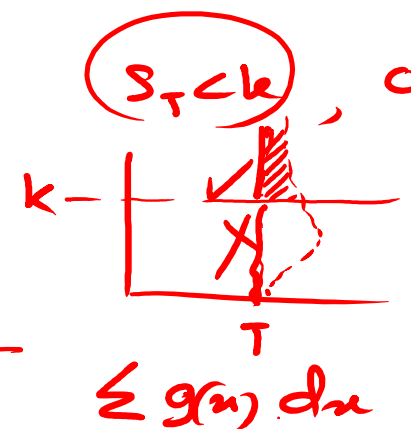
Call Price under GBM (quadrature)

Since the density of stock price under Black-Scholes is lognormal which is in closed form, we can directly perform numerical integration. The price of a call option is given as

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) \cdot f(x) dx$$

$m \times n$

$$C = \mathbb{E}^{\mathbb{Q}}[e^{-rT}(S_T - K)^+] = \int_K^{\infty} e^{-rT}(S_T - K)^+ f_{S_T} dS_T$$



For numerical integration using trapezoidal rule, we can use this discrete sum

→ lognormal.dist(x, μ, σ²) FALSE

$X \sim \text{lognormal}(\mu, \sigma^2)$

$\ln X \sim N(\mu, \sigma^2)$

$$C \approx \sum_{j=1}^N e^{-rT}(S_j - K)^+ f_{S_j} w_j \Delta S_j$$

$$w_j = \begin{cases} 0.5, & j = 0, n \\ 1, & \text{otherwise} \end{cases}$$

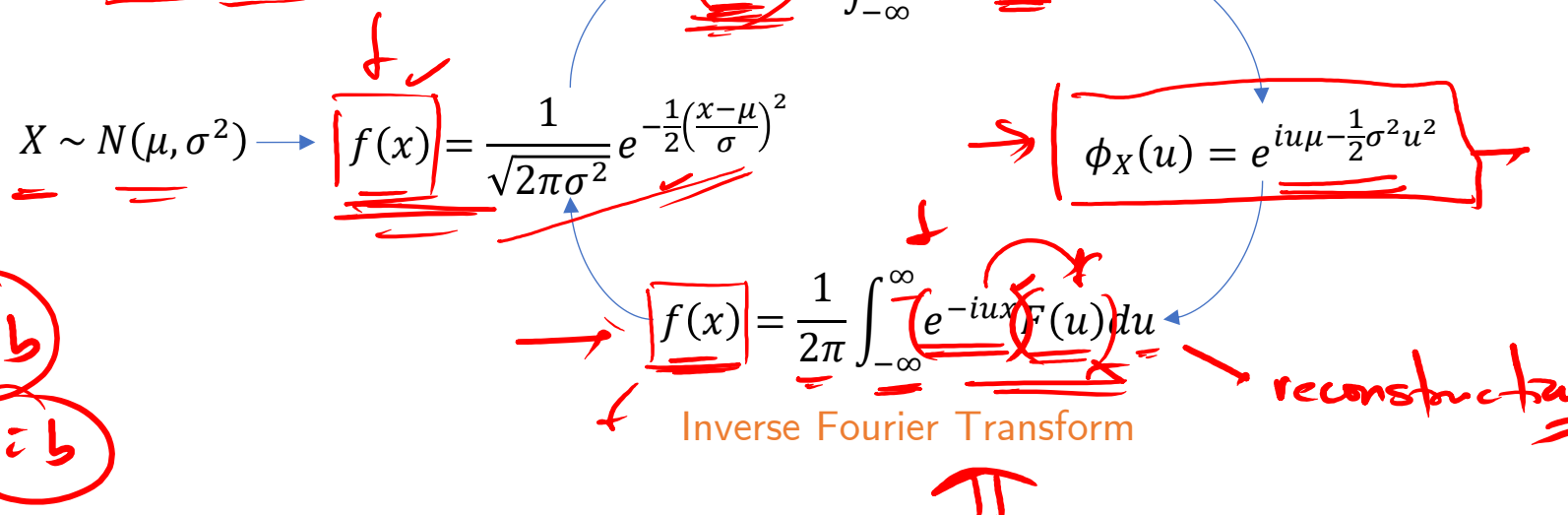
$S_T \sim \text{lognormal}(\quad)$

$\ln S_T \sim N(\ln S_0 + (r - \frac{1}{2}\sigma^2)T, \sigma^2 T)$

Fourier reconstruction of normal density (I)

Before we jump into option pricing using Fourier Transform, let's try to reconstruct Normal Density using Inverse Fourier from the characteristic function of the Normal Random. For a normal random, we have

$$\begin{aligned}
 (a+ib)(c+id) \\
 &= ac + aid + ibc - bd \\
 &= (ac - bd) + i(ad + bc)
 \end{aligned}$$



$$\begin{aligned}
 z &= a + ib \\
 \bar{z} &= a - ib
 \end{aligned}$$

Fourier reconstruction of normal density (II)

We can simplify the implementation by using the property that for a real valued function, the characteristic function is conjugate symmetric $\phi(-u) = \overline{\phi(u)}$, meaning the imaginary parts for u and $-u$ will have to cancel each other. Let's verify this

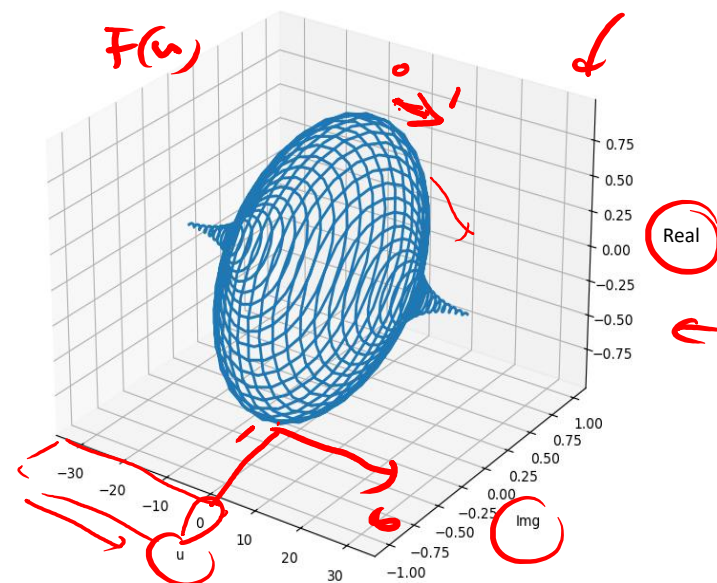
$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-iux} e^{iu\mu - \frac{1}{2}\sigma^2 u^2} du$$

$$f(x) = \frac{1}{2\pi} \sum e^{-\frac{1}{2}\sigma^2 u^2} \cdot e^{iu(\mu-x)} \Delta u$$

$e^{-\frac{1}{2}\sigma^2 u^2} (\cos(u(\mu-x)) - i \sin(u(\mu-x)))$

$e^{-\frac{1}{2}\sigma^2 u^2} (\cos(u(\mu-x)) + i \sin(u(\mu-x)))$

Handwritten notes: $e^{-i u(\mu-x)}$, $F(u)$, u_{min} , $-u$, u , u_{max} , $e^{-\frac{1}{2}\sigma^2 u^2}$



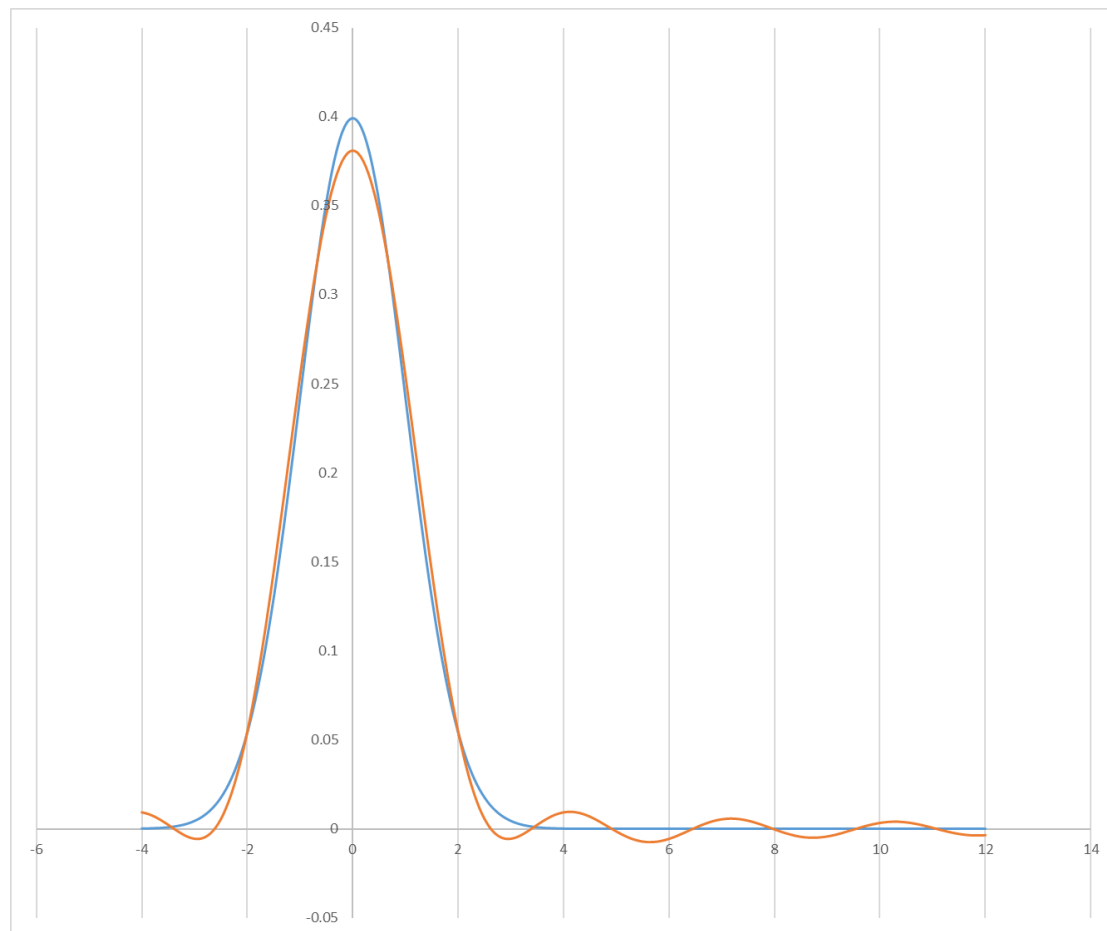
So instead of integrating u from $-\infty$ to ∞ , we can take 2 x the integrand of the real part and integrate from 0 to ∞ , i.e.

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \Re(e^{-\frac{1}{2}\sigma^2 u^2} \cdot e^{iu(\mu-x)}) du = \frac{1}{\pi} \int_0^{\infty} e^{-\frac{1}{2}\sigma^2 u^2} \cos(u(\mu-x)) du$$

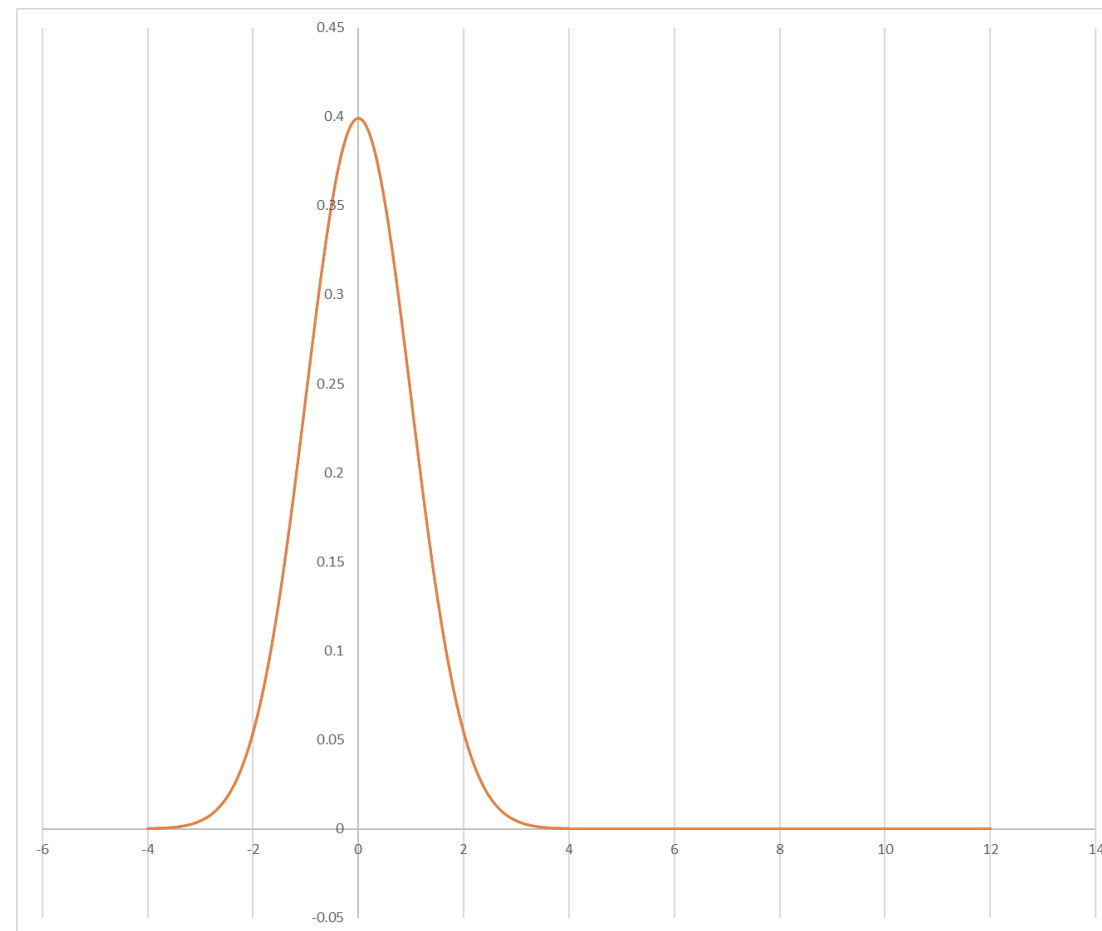
Fourier reconstruction of normal density (III)



$$\mu = 0, \sigma = 1, u_{max} = 2$$



$$\mu = 0, \sigma = 1, u_{max} = 5$$



Carr Madan Approach

Peter Carr and Dilip Madan published a paper named Option valuation using the fast Fourier transform in 1998 which was a revolutionary method for option pricing. Let

$$\underline{k} = \ln K \quad \underline{X_T} = \ln S_T$$

The European call as a function of log-strike is

$$\underline{C(k)} = e^{-rT} \mathbb{E} \left[(e^{X_T} - e^k)^+ \right]$$

$$\begin{array}{c} \textcircled{S_0} \begin{array}{l} \nearrow k_1 \\ \rightarrow k_2 \\ \searrow \vdots \\ \searrow k_n \end{array} \\ \textcircled{S_T - k}^+ \end{array}$$

Equivalently,

$$\underline{C(k)} = \underline{e^{-rT}} \int_k^\infty (e^x - e^k) \underline{q(x)} dx$$

where $q(x)$ is the density of X_T *risk-neutral*

Carr Madan Approach

A direct Fourier transform of $C(k)$ is not well behaved because call prices do not decay fast enough as $k \rightarrow [-\infty]$. Carr-Madan damping fixes this. Choose a damping parameter $\alpha > 0$.

Define the damped call value

$$c_\alpha(k) = e^{\alpha k} C(k)$$

Handwritten notes: A red arrow points from the text "Define the damped call value" to the equation. Above the equation, a red scribble is labeled u . To the right, a red circle contains S_0 and the text $u = \ln(k)$ is written next to it.

Define Fourier pair

$$\hat{c}_\alpha(v) = \int_{-\infty}^{\infty} e^{ivk} c_\alpha(k) dk$$

Handwritten note: A red circle around $\hat{c}_\alpha(v)$ is labeled (u) .

Fourier Transform

$$c_\alpha(k) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ivk} \hat{c}_\alpha(v) dv$$

Inverse Fourier Transform

Carr Madan Approach

Next, we derive the Fourier Transform of the damped call.

$$C(k) = e^{\alpha k} e^{-rT} \int_k^\infty (e^x - e^k) q(x) dx$$

Damped call

$$\hat{c}_\alpha(v) = \int_{-\infty}^\infty e^{ivk} e^{\alpha k} e^{-rT} \int_k^\infty (e^x - e^k) q(x) dx dk$$

Fourier Transform

Solving the integration leads to

$$\psi(v) = \hat{c}_\alpha(v) = e^{-rT} \frac{\phi_X(v - i(\alpha + 1))}{\alpha^2 + \alpha - v^2 + i(2\alpha + 1)v}$$

where ϕ_X is the CF. of $X_T = \ln S_T$

Fourier inversion formula

$$C(k) = e^{-\alpha k} c_\alpha(k) = \frac{e^{-\alpha k}}{\pi} \int_0^\infty \Re [e^{-ivk} \hat{c}_\alpha(v)] dv$$

$$C(k) = \frac{e^{-\alpha k}}{\pi} \int_0^\infty \Re [e^{-ivk} \psi(v)] dv$$

$$\psi(v) = \hat{c}_\alpha(v)$$

Carr Madan Approach

We can specify the discrete implementation of the integral using trapezoidal rule

Take a grid in the frequency space

$$(\Delta v) = \eta \quad \checkmark$$

$$v_j = j\eta, \quad j = 0, 1, \dots, N-1,$$

Handwritten: 0, η, 2η, 3η

where η is the frequency spacing. Using trapezoid weights

$$C_N(k) = \frac{e^{-\alpha k}}{\pi} \sum_{j=0}^{N-1} w_j \Re \left[e^{-i v_j k} \psi(v_j) \right] \eta$$

Handwritten: A scribbled-out equation to the left of the main formula.

This is often called a **discrete Fourier transform** implementation because the continuous Fourier inversion integral is replaced by a finite weighted sum over discrete frequencies. For one strike, it looks like quadrature. For many strikes, the same structure becomes a matrix-vector DFT.

For implementation, one can use $\alpha \in [0.75, 2]$, $N = 2^{15}$, $\eta = 0.01$

Handwritten: 1.5, FFT, and other annotations.

Carr Madan Approach



We need the call price for an array of strike prices for which we will have to use a DFT matrix. Now, if there are N points in the frequency domain and M points in the strike domain, our DFT matrix will be $\mathcal{O}(NM)$ which is very slow. In practice, we use FFT (Fast Fourier Transform) which requires a specialized grid structure.

Discretize the frequency as

$$v_j = j\eta, \quad j = 0, 1, \dots, N-1$$

Discretize the log-strike domain as

$$k_m = k_{\min} + m\lambda, \quad m = 0, 1, \dots, N-1$$

Also, the spacing between frequency and strike domain must satisfy the constraint

$$\lambda\eta = \frac{2\pi}{N}$$

The N is chosen to be a power of 2 (e.g., 2^{15}). The symmetry of this special grid structure allows FFT to reuse some calculations which improves the overall run time substantially. The FFT has an order of $\mathcal{O}(N \log N)$